

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Title: Outcomes of Nivolumab in Combination With Relatlimab in Patients With Advanced Melanoma in the Flatiron Database **Official Study Title:** Real World Outcomes of Nivolumab + Relatlimab in Patients With Advanced Melanoma in the Flatiron Database
Preferred UMLS Name: Nivolumab and Relatlimab for Advanced Melanoma **Protocol Number:** CA224-1098 **NCT Number:** NCT07079644
Posting ID: 181609042816260541 **Sponsor / Company Name:** Bristol Myers Squibb **Investigational Product:** nivolumab (Trade Names: Opdualag, Opdivo | ATC Code: L01FF01) in combination with Relatlimab (Biological); Comparators: Nivolumab + Ipilimumab, Immunotherapy Monotherapy, BRAF/MEK inhibitors **EMA Product Information:** [Opdivo EPAR](#), [Opdualag EPAR](#) **Phase of Development:** Not Applicable (Observational / Real-World Evidence) **Trial Status:** COMPLETED **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the real-world clinical outcomes of individuals diagnosed with advanced melanoma who were treated with Nivolumab in combination with Relatlimab, compared to other therapies (Nivolumab + Ipilimumab, immunotherapy monotherapy, or BRAF/MEK inhibitors). **Secondary Objectives:** To evaluate real-world overall survival (rwOS), real-world progression-free survival (rwPFS), time to next treatment, and the safety/tolerability of these treatments in routine clinical practice.

2.2 Background & Rationale To assess the real-world effectiveness of the fixed-dose combination of nivolumab and relatlimab (anti-PD-1 and anti-LAG-3) versus standard-of-care alternatives in advanced melanoma outside of controlled clinical trials. Utilizing the Flatiron Health database, this study provides vital insights into how these therapies perform in diverse, unselected patient populations within real-world clinical settings.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Retrospective Cohort Study) **Control Type:** Active Comparator (Nivolumab + Ipilimumab, Immunotherapy monotherapy, BRAF/MEK inhibitors) **Blinding:** Open-label / Unblinded (Database analysis) **Randomization:** N/A (Observational)

3.2 Planned Enrollment & Duration Targeted Enrollment: 678 adult patients **Number of Sites:** N/A (Extracted from the Flatiron electronic health records network) **Trial Status:** COMPLETED

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants in the advanced melanoma cohort. * Participants included in the cohort by Flatiron criteria of: cutaneous melanoma with pathologic stages III or IV, either at initial diagnosis or via local or distant recurrence, and at least two clinic encounters evident in the database. 2. Participants receiving Nivolumab + Relatlimab between March 18, 2022, and September 30, 2024, or other therapies (Nivolumab + Ipilimumab, immunotherapy monotherapy, BRAF/MEK inhibitors) between March 18, 2022, and November 30, 2024 (or date of the most recent data cut off). 3. Participants ≥ 18 years old on the index date. 4. Participants with ≥ 1 month of medical data after the index date (inclusive), unless the patient died within this period.

4.2 Exclusion Criteria 1. Participants with a diagnosis of other primary cancers prior to the index date. 2. Participants receiving medication in a clinical trial at any time prior to, on, or after the index date.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Real-world administration of Nivolumab + Relatlimab, Nivolumab + Ipilimumab, immunotherapy monotherapy, or BRAF/MEK inhibitors as prescribed by the treating physician in routine care. **Route of Administration:** Intravenous (for immunotherapies) or Oral (for targeted BRAF/MEK therapies). **Duration of Treatment:** Retrospectively analyzed from the treatment index date until discontinuation, death, or the end of the available follow-up period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for symptom management and comorbidities as per real-world practice. **Prohibited:** Concurrent participation in any interventional clinical trial.

6. Study Timeline & Visit Schedule

(Note: As a retrospective observational database study, timelines represent EHR data extraction points rather than prospective patient visits) | Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Pre-Index | Review of Flatiron EHR for eligibility, prior therapies, and baseline covariates | | **Baseline (Index Date)** | Day 0 |

Initiation of Nivolumab + Relatlimab or active comparator therapies | | **Observation** | Retrospective | Continuous EHR tracking for clinical response, progression, survival, and AEs | | **End of Study** | Data Cut-off | Final extraction of real-world outcomes up to November 30, 2024 (or date of the most recent data cut off) |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Demographics** 2. **Clinical characteristics:** Location of metastatic sites 3. **Clinical characteristics:** Lactate dehydrogenate (LDH) test results

7.2 Secondary & Exploratory Endpoints * Real-world clinical outcomes including real-world Overall Survival (rwOS) and real-world Progression-Free Survival (rwPFS) * Real-world Time to Next Treatment (rwTTNT) * Treatment Discontinuation Rates and Reasons for Discontinuation (e.g., toxicity vs. disease progression) * Rates of immune-mediated adverse events

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Retrospective extraction of documented immune-related adverse events and safety signals from physician notes and structured EHR fields. **Serious Adverse Events (SAEs):** Evaluated via hospitalization records, emergency department visits, and documented severe toxicities within the database. **Data Monitoring Committee (DMC):** N/A (Observational, retrospective database study).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Retrospective cohort of all eligible advanced melanoma patients treated with the specified regimens in the Flatiron network. **Sample Size Justification:** \$N = 678\$; represents the full available cohort of eligible patients meeting inclusion/exclusion criteria within the database timeframe.

Handling of Missing Data: Standard epidemiological methods for observational data, including inverse probability of treatment weighting (IPTW) or propensity score matching to account for baseline imbalances, and appropriate imputation or exclusion of missing covariates.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study is conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and strictly complies with all patient data privacy regulations (e.g., HIPAA de-

identification standards). **Monitoring Plan:** Automated and manual data abstraction quality checks within the Flatiron Health data processing pipeline. **Audit Trail:** Maintained securely by the database provider to ensure data provenance and traceability while preserving patient anonymity.

11. Study Identifiers & Governance

Primary ID: CA224-1098 **Registry Number:** NCT07079644 **Posting ID:** 181609042816260541 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** Outcomes of Nivolumab in Combination With Relatlimab in Patients With Advanced Melanoma in the Flatiron Database **Official Regulatory Title:** Real World Outcomes of Nivolumab + Relatlimab in Patients With Advanced Melanoma in the Flatiron Database

12. Clinical Framework (Advanced Melanoma Specific)

Disease Area / Primary Condition: Advanced Melanoma **Preferred Name:** Advanced Melanoma **Disease Definition:** Melanoma that has spread extensively to other anatomic sites or is no longer responding to treatment. **Semantic Type:** Neoplastic Process **MeSH Code:** N/A **SNOMED ID:** 186715002 **SNOMED Hierarchy:** 254830009|Malignant melanoma|Malignant melanoma (disorder) **Diagnosis Threshold:** Histologically confirmed advanced (unresectable or metastatic) melanoma (pathologic stages III or IV) **Medical Methodology:** Immune Checkpoint Inhibition (anti-PD-1 / anti-LAG-3, anti-CTLA-4) vs. Targeted Therapy (BRAF/MEK inhibitors) **Optimization Target:** Improvement in real-world progression-free and overall survival

13. Study Procedures & Methodology

Management Pathway: Real-world standard of care for advanced melanoma **Education Protocol (HCP):** N/A (Retrospective EHR Study) **Education Protocol (Patient):** N/A (Retrospective EHR Study) **Quality Audit Mechanism:** Source data verification, abstraction auditing, and quality control metrics executed by Flatiron Health

14. Observation & Follow-Up Schedule

Total Duration: Retrospective extraction from Index Date (on or after March 18, 2022) to data cut-off (up to November 30, 2024, or most recent data cut off) **Onsite Visit Cadence:** N/A (EHR abstraction) **Remote Monitoring Frequency:** N/A **Checklist Review Interval:** N/A

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					